COURSE TITLE : HOSPITAL SYSTEMS
COURSE CODE : 6242
COURSE CATEGORY : A
PERIODS/WEEK : 6
PERIODS/SEMESTER : 90
CREDITS : 6

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Hospital Organization, Architecture & Electrical power supply systems	23
2	Gas supply systems and other Services in Hospitals	22
3	Glucose monitoring, foetal monitoring and Audiometer	23
4	Biotelemetry and telemedicine.	22
TOTAL		90

Course General Outcome:

MODULE	G.O.	ON COMPLETION OF THE STUDY OF THIS MODULE THE STUDENT WILL BE ABLE
1	1	To understand Hospital Organization and Architecture
	2	To understand Electrical power supply systems in Hospitals
2	3	To understand Gas supply systems in Hospitals
	4	To comprehend sterilization techniques, incinerators and theatre lighting systems
3	5	To understand Glucose monitoring
	6	To understand foetal monitoring and Audiometer
4	7	To understand biotelemetry
	8	To understand telemedicine

G.O.: General Outcome

On completion of the study the student will be able

MODULE I HOSPITAL ORGANIZATION, ARCHITECTURE & ELECTRICAL POWER SUPPLY SYSTEMS

1.1.0 To understand Hospital Organization and Architecture

1.1.1 To define Bio-engineering, bio –medical Engineering, Clinical Engineering and Hospital Engineering
1.1.2 To describe the duties and responsibilities of Biomedical engineer in a hospital
1.1.3 To explain Hospital Organization chart
1.1.4 To describe the Architecture and Planning of a Hospital
1.1.5 To describe design of various wards and ICU

1.2.0 To understand Electrical power supply systems in Hospitals

1.2.1 To illustrate the wiring details needed in a Hospital
1.2.2 To explain the working of power supply systems

1.2.3 To state the importance of uninterrupted power supply in Hospital
1.2.4 To list the importance of voltage stabilizers
1.2.5 To explain circuit breakers
1.2.6 To explain protective relays
1.2.7 To explain surge protector
1.2.8 To describe Electromagnetic interference filter
1.2.9 To explain the physiological effect of electrical currents, micro shock and macro shock.

MODULE II GAS SUPPLY SYSTEMS AND OTHER SERVICES IN HOSPITALS

2.1.0 To understand Gas supply systems in Hospitals

2.1.1 To explain the operation of Centralised gas supply system in hospitals.
2.1.2 To explain the operation of Centralised vacuum supply system in hospitals
2.1.3 To explain the operation of liquid oxygen plant in a hospital.
2.1.4 To list the safety requirements of gas supply systems in hospitals

2.2.0 To comprehend sterilization techniques, incinerators and theatre lighting systems

2.2.1 To define Sterilization
2.2.2 To classify different types of sterilization
2.2.3 To describe single and double chamber autoclave
2.2.4 To describe ethylene oxide Gas sterilization
2.2.5 To state the importance of incinerator.
2.2.6 To list the characteristics of theatre lighting systems.

MODULE III GLUCOSE MONITORING, FOETAL MONITORING AND AUDIOMETER

3.1.0 To understand Glucose monitoring

3.1.1 To define biosensor
3.1.2 To describe membrane electrodes
3.1.3 To describe blood glucose sensor using ISFET
3.1.4 To explain construction of an enzyme electrode
3.1.5 To explain the principle and operation of glucometers.

3.2.0 To understand foetal monitoring and Audiometer

3.2.1 To explain the principle and operation of incubators.
3.2.2 To explain the principle and operation of Doppler foetal monitor.
3.2.3 To explain the principle and operation of pure tone audiometry.
3.2.4 To explain the principle and operation of speech audiometry.
3.2.5 To explain the block diagram of an audiometer.

MODULE IV BIOTELEMETRY AND TELEMEDICINE.

4.1.0 To understand biotelemetry

4.1.1 To explain the principle and components of bio –telemetry system.
4.1.2 To describe the physiological parameters adaptable to bio-telemetry.
4.1.3 To explain single channel telemetry system .
4.1.4 To explain multichannel telemetry system.

4.2.0 To understand telemedicine

4.2.1 To define telemedicine.
4.2.2 To list the applications of telemedicine.

4.2.3 To explain the basic parts of a teleradiography system.
4.2.4 To explain the basics of tele-conferencing and video conferencing

CONTENTS

MODULE I HOSPITAL ORGANIZATION AND ARCHITECTURE

Definition of Bio-engineering , bio –medical Engineering, Clinical Engineering and Hospital Engineering - Duties and responsibilities of Biomedical engineer in a hospital - Hospital Organization chart- Architecture and Planning of a Hospital - Design of various wards and ICU.

Electrical power supply systems in Hospitals

Wiring details needed in a Hospital - Working of power supply systems-Importance of uninterrupted power supply in Hospital and voltage stabilizers-Circuit breakers-Protective relays-Surge protector-Electromagnetic interference filter-Physiological effect of electrical currents, micro shock and macro shock.

MODULE II GAS SUPPLY SYSTEMS IN HOSPITALS

Operation of Centralised gas supply system and centralised vacuum supply system in hospitals-operation of liquid oxygen plant in a hospital- safety requirements of gas supply systems in hospitals

Sterilization techniques, incinerators and theatre lighting systems

Definition of Sterilization- Types of sterilization- Single and double chamber autoclave- Ethylene oxide Gas sterilization- Importance of incinerator- Characteristics of theatre lighting systems.

MODULE III GLUCOSE MONITORING

Definition of biosensor- Membrane electrodes- Blood glucose sensor using ISFET- Construction of an enzyme electrode- Principle and operation of glucometers.

Foetal monitoring and Audiometer

Principle and operation of incubators, Doppler foetal monitor, pure tone audiometry, speech audiometry- Block diagram of an audiometer.

MODULE IV BIOTELEMETRY

Principle and components of bio –telemetry system- physiological parameters adaptable to bio-telemetry- single channel telemetry system - Multichannel telemetry system.

Telemedicine

Definition of telemedicine-applications of telemedicine- Parts of a teleradiography system-Tele-conferencing and video conferencing

TEXT BOOK

1.Handbook of Biomedical Instrumentation- RS Khandpur- Tata McGraw-Hill Pub., 2006t distance.
lecytology: Applications, Telecardiology: requirements, portable solutiopREPACOOOREDCCCAACOHJOJNKns

2.Hospitals Planning,Design and management-GD Kunders,S Gopinath,A Katakam-Tata McGrawHill Publishing Company, 1998

REFERENCES

1. Clinical Engineering-C A Caceres-Academic press Newyork 1977
2. Standard hand book of Biomedical Engineering & Design- Kutz Myer-McGraw Hill-2002
3. Electric power substation engineering- John Douglas McDonald-CRC Press 2003
4. Applied biosensors-D L Wise-Butterworth Publishers London-1989
5. Automatic ventilation of lung-Mushin-Black well-1980
6. Principles of Applied Biomedical Instrumentation-Geddes & Baker-wiley 1989